THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SOCIAL SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 4: **Atmosphere and Climate (Geography)** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a news bulletin: 'An aurora lit the skies over Norway.' Which layer is responsible? 1

(a) Mesosphere **(b)** Thermosphere **(c)** Stratosphere **(d)** Troposphere

2. Assertion (A): The south-west monsoon brings most of India's rain. Reason (R): Moist winds blow from sea to low-pressure land. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. A fisherman sails out at night using the 1

(a) monsoon **(b)** sea breeze **(c)** land breeze **(d)** storm

4. Assertion (A): Temperature falls with altitude in the troposphere. Reason (R): The stratosphere is heated by absorption of UV by ozone. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): Greenhouse gases cool the Earth. Reason (R): They trap heat in the atmosphere. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. If the temperature in a layer reaches highest at about 700 km and radio waves reflect there, it is 1

(a) stratosphere **(b)** troposphere **(c)** thermosphere **(d)** mesosphere

7. Assertion (A): Planting trees reduces carbon footprint. Reason (R): Trees absorb carbon dioxide. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. Assertion (A): The north-east monsoon brings rain to Tamil Nadu. Reason (R): The winds pick up moisture over the Bay of Bengal. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** Aeroplanes prefer to fly in the lower stratosphere. 1
Reason (R): The stratosphere is calm and free from clouds.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Temperature falls with altitude in the troposphere. 1
Reason (R): The stratosphere is heated by the ozone layer absorbing ultraviolet radiation.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does temperature rise in the thermosphere though air is thin? 2

12. Why might a forecast of normal monsoon still bring local floods? 2

13. Why is NMM important for farmers? 2

14. Why are cities with high pollution at risk during heat waves? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. How can planting trees along rivers reduce flood damage? 3

16. Explain why a calm, cloud-free layer is useful for aircraft but not for weather forecasts. 3

17. Explain why a monsoon forecast is more useful when combined with local data. 3

OR

Explain how a delayed monsoon affects farmers and the economy. 3

18. Explain how the IMD's four seasons differ from the traditional six ritus. 3

OR

Explain how sea breezes help the fishing and tourism industries of a coastal town. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A teacher says: 'Weather happens in the lowest layer, which has an average height of about 12 km. Above it lies a layer containing ozone that is calm and cloud-free. Most meteors burn up in a higher layer. In the layer at about 700 km, radio waves are reflected back and auroras appear.' 5
(i) Name the lowest layer of the atmosphere.
(ii) Which layer contains the ozone layer?
(iii) In which layer do most meteors burn up?
(iv) Name the layer where radio waves are reflected and auroras occur, and state why jet aircraft prefer the lower stratosphere.

20. Discuss how climate change is a global challenge and how individuals can respond. 5

OR

Compare weather and climate and explain how each influences human life with examples. 5

END OF QUESTION PAPER — Total: 40 Marks